

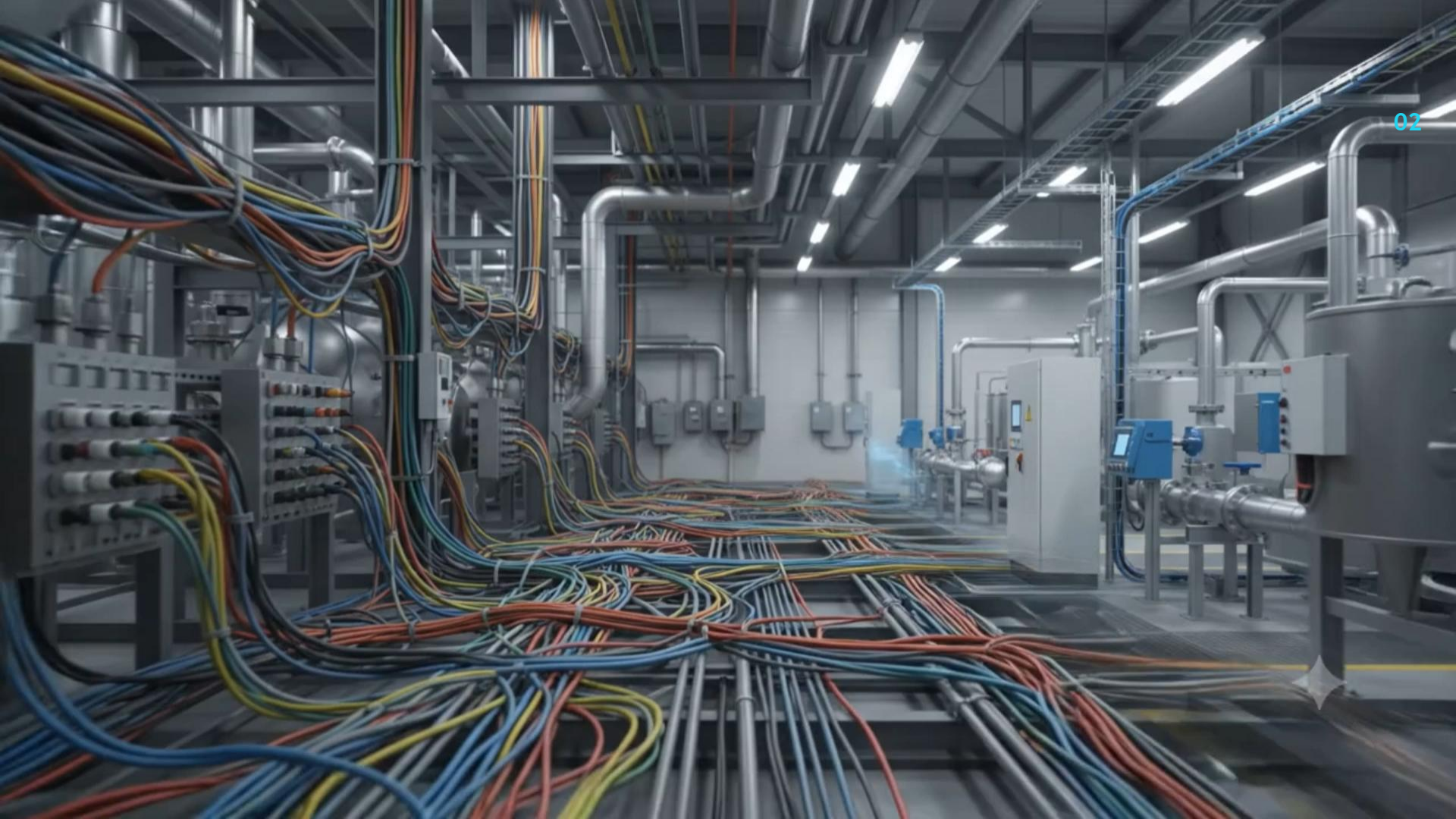
IEEE 802.3cg • 10BASE-T1L

Advanced Physical Layer Ethernet Ethernet-APL

Bringing Ethernet into Hazardous Process Industries

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Why Do We Need Ethernet-APL?

03

THE PROBLEM – legacy field communication

Technology	Limitation
4–20 mA	One value only
HART	Very slow
PROFIBUS PA	Limited bandwidth

INDUSTRY 4.0 REQUIRES

- ✓ Diagnostics
- ✓ Cloud connectivity
- ✓ Predictive maintenance
- ✓ Digital twins

4-20 mA



HART



PROFIBUS PA

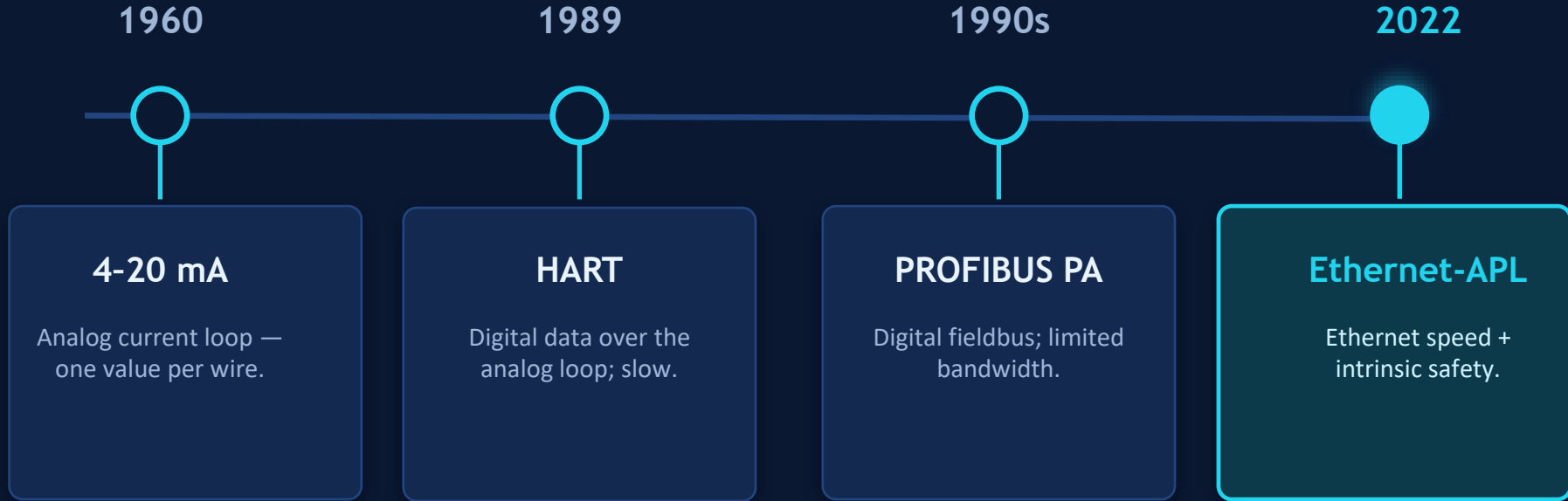


Ethernet-APL

evolution toward Ethernet in the field

Evolution of Industrial Communication

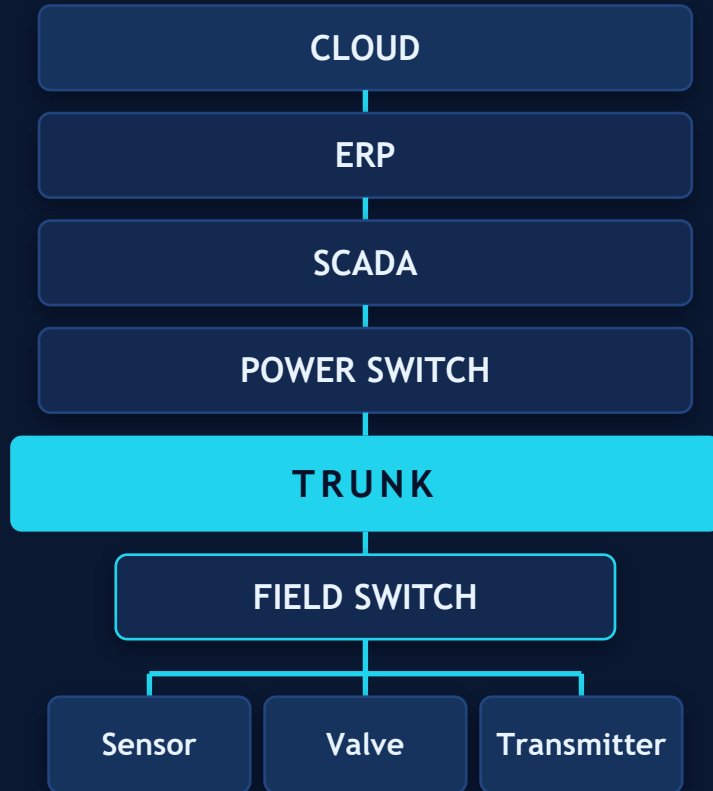
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Key message: Ethernet-APL is the first technology that combines Ethernet speed with intrinsic safety.

Ethernet-APL Architecture

05



TRUNK

Backbone segment
1000 m · 10 Mbps

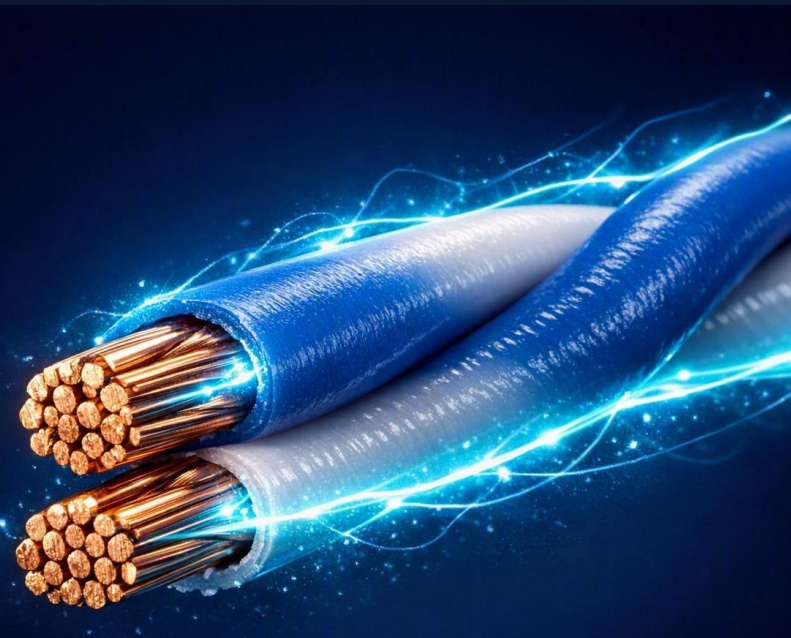
SPUR

Branch to field devices
200 m · Zone 0 capable

Hierarchy from enterprise cloud down to intrinsically-safe field devices.

How Ethernet-APL Works

06



Single Pair Ethernet — power + data on one twisted pair

IEEE 802.3cg

10BASE-T1L physical layer

- ✓ Single Pair Ethernet (one twisted pair)
- ✓ 10 Mbps data rate
- ✓ Full-duplex communication
- ✓ Up to 1000 m reach
- ✓ Power + data on the same cable

Why Hazardous Areas Are Difficult

07

ZONE CLASSIFICATION

Zone	Meaning
Zone 0	Always explosive
Zone 1	Sometimes explosive
Zone 2	Rarely explosive

Hazardous zones are ranked by how often an explosive atmosphere is present.

✗ Standard Ethernet can't operate safely

- Too much energy
- Not intrinsically safe
- RJ45 connector unsuitable

✓ Ethernet-APL solves this

Energy-limited single pair, intrinsically safe by design, purpose-built for Zone 0–2.

Intrinsic Safety — 2-WISE

08

2-Wire Intrinsically Safe Ethernet

HOW IT WORKS

1

Based on IEC TS 60079-47
the 2-WISE technical specification

2

Limits voltage and current
energy kept below ignition threshold

3

Prevents ignition
no spark can release enough energy

High Energy



Explosion



Low Energy



Safe
Operation

ATEX

IECEX

Zone 0 certified

Comparison at a Glance

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Feature	HART	PROFIBUS PA	Ethernet-APL
Speed	1.2 kbps	31.25 kbps	10 Mbps
Ethernet	No	No	Yes
Cloud Ready	No	Limited	Yes
Industry 4.0	Poor	Medium	Excellent

Ethernet-APL outperforms legacy fieldbuses on every dimension that matters for Industry 4.0.

Industry 4.0 Impact

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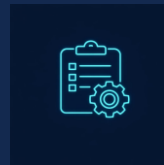
Ethernet-APL unlocks the connected, data-rich field that Industry 4.0 depends on.



Digital Twin



OPC-UA



Asset
Management



Predictive
Maintenance



Cloud Analytics



AI at Field
Devices



Future Technologies

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100BASE-T1L

01

10× faster single-pair Ethernet

OPC-UA TSN

02

Deterministic real-time Ethernet

5G + Ethernet-APL

03

Hybrid wired & wireless networks

Edge AI

04

Local intelligence at the device

Conclusion

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1

Ethernet-APL brings Ethernet directly into hazardous areas.

2

Delivers 10 Mbps over 1000 m using one cable pair.

3

A foundation technology for Industry 4.0 and IIoT.

We are grateful for your attention !

Any Questions?

Q1 Why not simply use Wi-Fi or 5G?

Q2 Why is intrinsic safety necessary?

Q3 How does it differ from standard Ethernet?

Q4 Why is 10 Mbps sufficient?

Q5 What role does OPC-UA play?

Q6 Can it fully replace PROFIBUS PA?